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GENE 404 Final Report

Design of an Ethanol-Liquid Oxygen Rocket Engine with Thrust Vector Control

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2 Short Abstract

Humanity's future is increasingly tied to our ability to access space. In pursuit of the technologies that make economical space access possible, the goal of this project was to build a small-scale rocket engine powered by ethanol and liquid oxygen. The engine demonstrated thrust vector control which allows for the movement of the engine while it is firing. This laid the groundwork for the development of a hopper vehicle that can take off and land autonomously.

3 Long Abstract

The commercialization of rocket launches in recent years by SpaceX has significantly reduced the cost of space access. A major factor in the decrease of the cost of launches comes from the ability to land and reuse rocket boosters [1]. Developing the technologies required to propulsively land rocket stages domestically will enable increased access to space from Canada.

The primary goal of this project was to develop and test a non-flight rocket engine using liquid oxygen, a cryogenic oxidizer, and ethanol fuel. A copper combustion chamber was chosen to act as a heatsink, allowing for a targeted 10 second burn time without complicated cooling systems. The engine was capable of gimbaling using thrust vector control (TVC). This means it could precisely point the direction of thrust while firing to control a rocket stage when it is launching or descending to a landing pad. Supporting these were a custom cryogenic fluid feed system and control algorithm for precise engine position control.

In the future, capstones and members of the Waterloo Rocketry team will be able to expand the project such as with thrust throttling, regenerative cooling and eventually a full flight vehicle. This project will help develop the expertise of these technologies domestically in Canada and contribute to the following United Nations Sustainable Development goals: Industry, Innovation and Infrastructure, Responsible Consumption and Production, and Quality Education.

4 Executive Summary

Reusable rockets will lower the cost of space access. This report details the development of a small scale ground engine which demonstrates many of the key technologies required for a small-scale “hopper” reusable vehicle. The engine features thrust vector control and liquid oxygen as a propellant; two technologies which are critical for future self-landing booster. A copper heatsink engine was designed to produce 500 N of thrust and burn for 10 seconds. The engine used ethanol and cryogenic liquid oxygen as its fuel and oxidizer respectively. Liquid oxygen was chosen because of its efficiency and wide use in industry. The thrust vector control and gimbal system were designed to meet parameters based on a Simulink model of a small-scale hopper vehicle. A ring-style gimbal was used because it is relatively simple to manufacture and allows fluid connections through its center. A custom fluid feed system was built to provide propellants to the engine at up to 400 psi. New operational and safety procedures were developed to work with cryogenic propellants. Ground support equipment was adapted from the Waterloo Rocketry design team, including a custom test stand, electrical ground support equipment, a test site and a flame diverter. A custom printed circuit board was designed to run a control loop and drive the thrust vector control actuators. Software was used to create an automatic valve actuation sequence for firing the engine remotely.

Two cold flows, one TVC “wiggle” test and two unsuccessful static fire tests were conducted. From these tests, it was determined that the feed system and engine were able to handle cryogenic propellants at the desired pressures and flow rates. The thrust vector control system also met all of its performance requirements. Cryogenic operations procedures were verified and improved over the testing campaign. Engine performance was not able to be tested due to issues with the static fire test, however a second static fire attempt is planned to characterize the engine.

It is recommended that the Waterloo Rocketry student design team proceed with further development of thrust vector control systems and use of cryogenic propellant in line with their long-term goals. However, doing so must be done in a safe manner and with the proper regulatory approvals.

5 Introduction

Access to space has resulted in technological and societal benefits, from the development of various new technologies to a greater understanding of our own biology. Currently only a few companies have the ability to access space, and even fewer have the ability to do this relatively cheaply. As a result, this project aims to design some of the technologies required to create accessible alternatives. This includes thrust vector control and a liquid bipropellant rocket engine.

The purpose of this report is to describe the final design and all the research and development that went into it. In addition, the tests that were used to characterize this design are also described along with their results.

Specifically, the report is broken down into the following sections.

Section 6 - Problem Context The main motivations for the project are described, along with the current state of the art.

Section 7 - Review of the literature All the literature reviewed for this project.

Section 8 - Research Questions and Objectives Objectives of this project and the standards by which they are measured, including descriptions of the testing methodology.

Section 9 - Methodology Design details of each subsystem, including requirements, decisions, and other factors that influenced the final product.

Section 10 - Results The various tests performed during the projects are described along with their results.

Section 11 - Discussion A summary of the results and a comparison with the original goals.

Section 12 - Recommendations Directions and advice to those who continue the project, in particular the Waterloo Rocketry student design team.

6 Problem Context

The monetary, material and environmental costs of achieving orbit limit humanity's ability to expand industry and research into space. Historically, rockets have been single-use: discarded into the ocean as scrap after every launch. This is the simplest approach and is quite practical when rockets were government-funded projects with incredible budgets. To this day many launch companies continue to follow this methodology, and find success in various government-contracted missions [2].

However, as demand for access to space has increased in recent years, especially from the private sector, there has been pressure to reduce launch costs. More applications for satellites continue to arise, from GPS and global internet to earth observation and space mining. The industry has responded by becoming commercialized, with companies trying to break into the market by using

new technology to make launches cheaper. The most significant technological breakthrough in this effort has been the reuse of rocket boosters, as demonstrated primarily by SpaceX [3].

By designing their rocket boosters to self-land in one piece, SpaceX is able to refurbish and re-fly their boosters many times. This approach has shattered the cost floor for launches to orbit, with SpaceX’s Falcon 9 and Falcon Heavy rockets coming in at fractions of the price of competitors [1], as shown in Figure 1. While other companies are working to develop similar reusable technology, propulsive self-landing technology is currently extremely proprietary which severely limits its adoption and thus economical access to space.

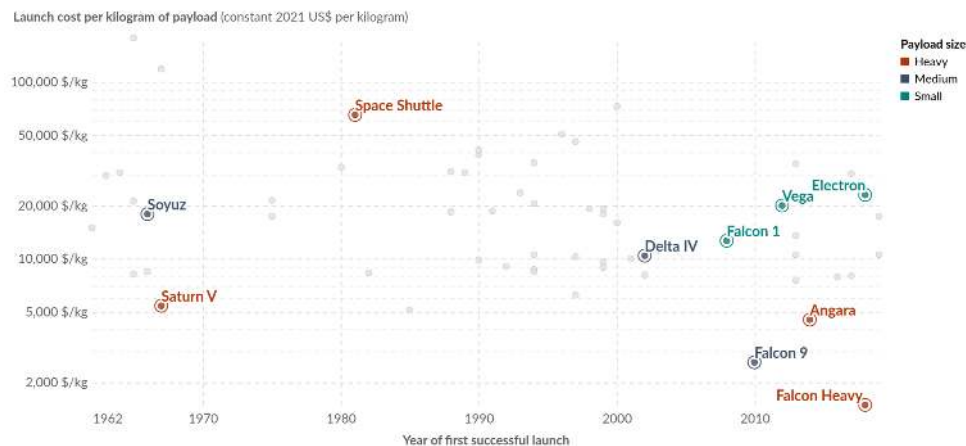


Figure 1: Cost of space launches to low earth orbit. [1]

It is envisioned that, if the technology required for reusable self-landing rocket boosters was developed openly, more companies would be able to adopt it and compete. This would both push self-landing technology forward further and serve to reduce the costs to access space, unlocking future industrial and research applications.

7 Literature Review

The textbook *Design of Liquid Propellant Rocket Engines Second Edition* by Huzel and Huang, published on the NASA Technical Reports Server [4], has been very useful during the design of the LOX-ethanol engine. The book contains useful information for designing a liquid propellant rocket engine such as different design parameters, various formulas, tables of empirical and precalculated values, properties of common fuel and propellants, and design steps. The engine design of this capstone project heavily references this textbook.

Liquid Rocket Engine Injectors by Gill and Nurick was another useful resource, also published on the NASA Technical Reports Server [5]. This work includes many design considerations for rocket engine injectors. In particular, it lists out most of the existing injector designs with their

advantages, disadvantages, and typical use cases. This report provided grounding for the choice of injector for this capstone project.

Finally, the engine design in *Design and Testing of a Liquid Nitrous Oxide and Ethanol Fueled Rocket Engine* by Youngblood [6] thoroughly documented the development and testing of a small-scale LOX-ethanol engine very similar to the goal of this project. This work provided a good reference for the design of this project, specifically the use of copper as the combustion chamber material for its heatsink capability.

8 Research Questions and Objectives

The primary goal of this project was to develop and demonstrate two key technologies required for self-landing rockets: thrust vector control and cryogenic propellants. These technologies were demonstrated by designing and testing a small ground engine, which removes the complexity of designing for flight weight or integrating the technologies onto a full rocket.

Thrust vector control is the ability to precisely control the direction of thrust of a rocket engine produces while it is firing. This is critical to enabling self-landing rockets, as during descent the rocket must be precisely balanced like an inverted pendulum by varying the thrust direction. Validating the ability to direct an engine’s thrust while firing statically on the ground is an important stepping stone towards doing so on a flight vehicle.

Another key requirement of reusable rockets is that they must be efficient, which requires the use of a liquid rocket engine using cryogenic propellant. Most amateur liquid rockets use nitrous oxide as their oxidizer since it is relatively simple to work with, but in industry almost all rockets use liquid oxygen instead as it is much more efficient [4]. The successful design of a rocket engine using liquid oxygen would aid in opening up its superior performance at the amateur level.

A variety of tests to validate the technologies were planned. Thrust vector control can be tested by performing a “wiggle”, moving the engine without firing it to validate the mechanical systems and control algorithms, before moving up to an integrated test with the engine firing. Cryogenic performance can be validated through “cold flows” that use inert cryogenic liquids in place of liquid oxygen to validate the system’s performance at low temperatures. Finally an integrated “static fire” can be performed by actually firing the engine while it is bolted down to measure thrust and other performance metrics.

9 Methodology

The design of the ground engine was split into multiple independent systems to allow parallel work while reducing integration requirements. Minimal sets of systems could then be assembled as they were ready to perform the various planned tests, allowing their designs to be proven out as quickly

as possible. Once all of the systems had been tested they could then all be assembled together for the integrated static fire test.

9.1 Requirements Generation

The requirements for this project originate from making a hopper vehicle. Some of the requirements come from the Lander Challenge [7], a competition that challenges student teams to build self-landing rocket boosters. This includes the minimum of 7 degrees of gimbal angling.

The initial requirements inspired a Simulink simulation which featured a simplified hopper. It was two cylindrical masses stacked on top of each other, the lower of which was constantly applying 500 N of force onto the upper. The lower mass could pivot up to 7 degrees in all directions and a PID loop ensured that the hopper remained upright. The maximum angular speed and acceleration obtained from these simulations became the requirements for the TVC system. Figure 2 and Figure 3 show the mechanical model used in Simulink, as well as a 3D view of the system. These simulations gave a required angular speed of 0.4 rad/s, an angular acceleration of 2 rad/s².

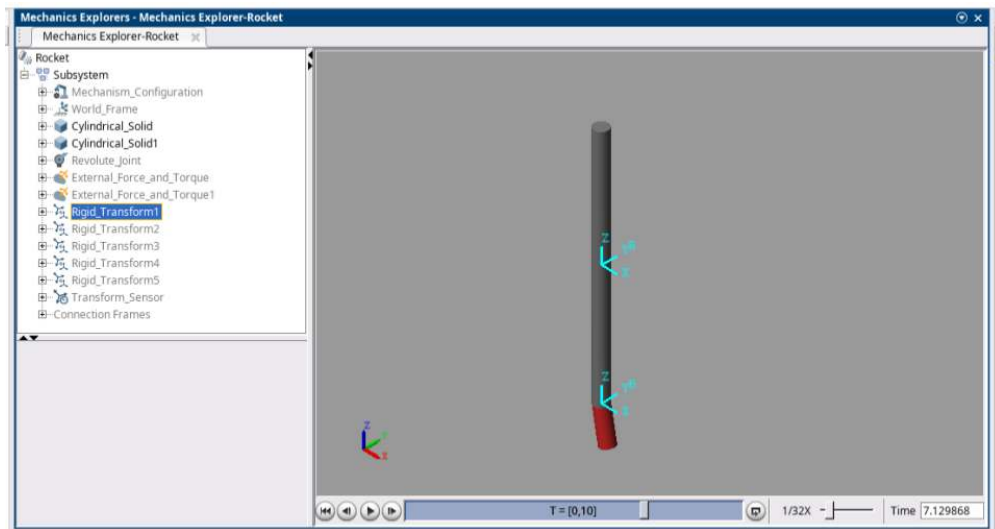


Figure 2: 3D view of simulink simulation used to derive requirements.

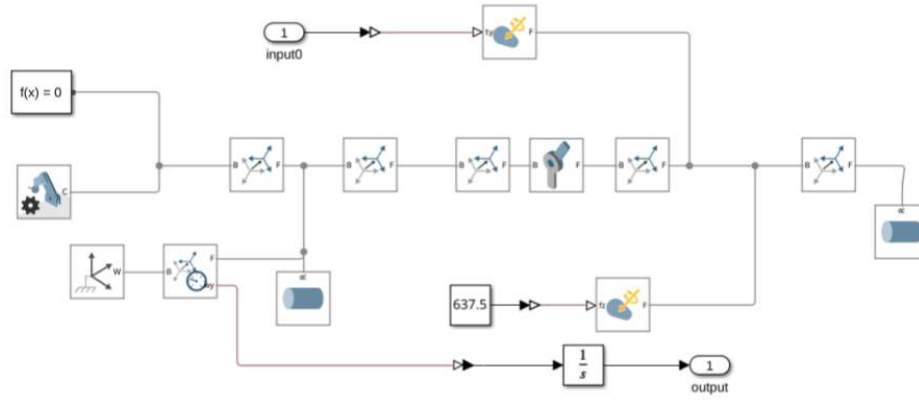


Figure 3: Mechanical model used to define the Simulink simulation.

In addition to the quantitative requirements, there were numerous other qualitative requirements. Principal among these were simplicity and scalability. The scope of the project was quite large as is, so numerous compromises were made to reduce the difficulty. This includes using a heat sink engine rather than developing regenerative cooling and using commercially available actuators rather than designing them from scratch.

9.2 Engine

The engine is responsible for mixing and combusting the propellants provided by the feed system to produce thrust. It is mounted to the gimbal, which allows it to rotate, and the rotation is performed by the thrust vector control system. It consists of two key components: the injector, which mixes the propellants, and the combustion chamber, which contains the combustion and has a nozzle to create thrust.

Initial requirements for the engine were set with the goal of keeping the design as simple and reliable as possible, rather than trying to optimize for efficiency, in order to manage the scope of the project. Liquid oxygen was selected as the oxidizer for its use in industry and to meet the project goal of using a cryogenic propellant. The fuel chosen was ethanol, as the Waterloo Rocketry team has experience using it and it is easy to obtain and work with being a liquid at room temperature with a low vapor pressure. The target thrust was chosen arbitrarily as 500 N, a low value meaning the engine and feed system could be small while remaining easy to fabricate. The target burn duration was chosen to be 10 seconds to allow sufficient time to demonstrate thrust vector control.

In keeping with the goal of simplicity, the cooling method of the combustion chamber (CC) was chosen to be a heatsink, meaning that the CC simply absorbs the heat of combustion rather than having any more complex form of active cooling. To limit the energy transferred into the CC, the combustion temperature needed to be kept low, which was achieved by selecting an oxidizer

to fuel ratio of 1 and a combustion chamber pressure of 200 psi. This leads to less efficient but low-temperature combustion. The key engine parameters are summarized in Table 1.

Table 1: Key engine parameters

Parameter	Value
Propellants	Liquid Oxygen / Ethanol
Thrust	500 N
Burn Duration	10 s
O/F	1
CC Pressure	200 psi
Propellant Flow Rate	238 g/s

The combustion chamber inner diameter was chosen to be 2.75 in, and the integral nozzle dimensions were determined using the Rocket Propulsion Analysis software [8]. Finite element analysis simulations were then performed using ANSYS to determine the required material and wall thickness for the combustion chamber, as shown in Figure 4. It was determined that a copper chamber was required to conduct the head loading fast enough to avoid melting, with an outer diameter of 4 in.

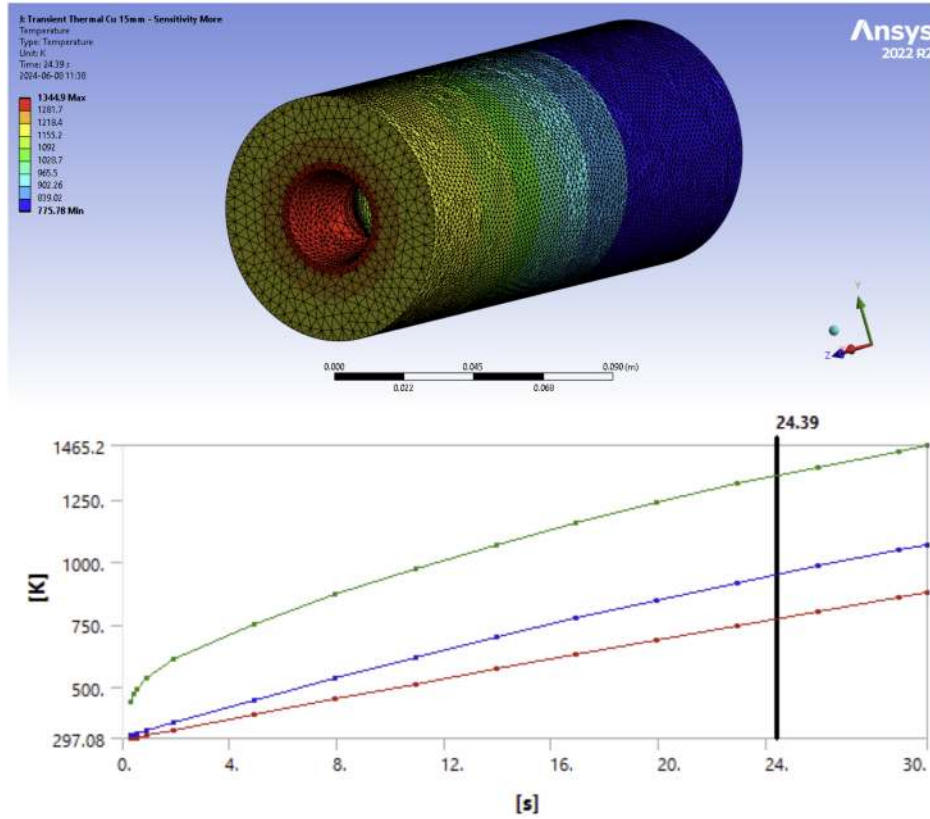


Figure 4: Simulation of CC temperature over time

The other main component of the engine is the injector. In keeping with the design goal of simplicity, an unlike impinging injector was selected. This type of injector directs streams of fuel and oxidizer at each other where they collide, mix, and combust. A 6-element pattern was selected for fairly even distribution inside the chamber, and the elements were then sized for a pressure drop of at least 25% of the chamber pressure (50 psi) and to line up with standard drill bit sizes. This resulted in 6x #59 drill (1.04 mm) holes for the oxidizer and 6x 3/64 in (1.18 mm) holes for the fuel. The elements were arranged in a radial pattern for simple manifolding. A cross-section of the combustion chamber and injector are presented in Figure 5.

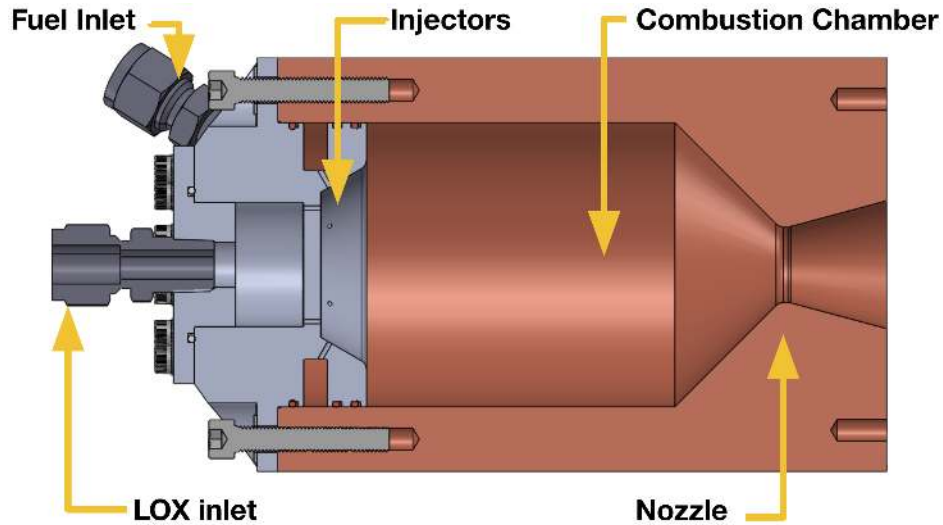


Figure 5: Cross section of the CC and injector

All sealing in the engine was performed using o-rings, which were selected for their design simplicity and robustness. Radial silicone o-rings were used to seal the combustion chamber and the fuel manifold, as they perform well at high temperatures. A PTFE o-ring in a face seal configuration was selected to seal the liquid oxygen inlet into the injector bulkhead. PTFE was selected for its material compatibility and ability to seal at low temperatures, and a face seal was chosen to allow the components to thermally contract without compromising the seal. The various components of the engine were attached using axial bolted joints, all designed following [9]. The seals and bolted joints were tested by attaching a plug to the end of the CC and pressurizing it with water.

9.3 Feed System

The feed system is a pressurized fluid system whose purpose is to deliver propellants to the engine at a specified flow rate and pressure. It consists of fluid systems components mounted on a welded steel frame. It is designed to fit through a standard sized door and has wheels for ease of transportation. Photos of the front and back of the assembly are shown in Figure 6.

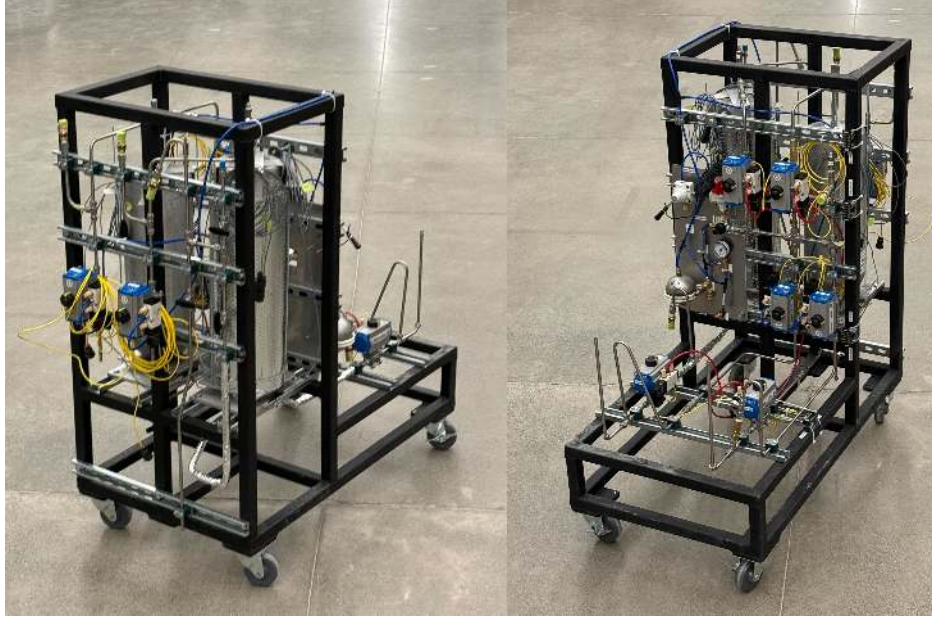


Figure 6: Feed system back (left), feed system front (right)

The feed system controls three different fluids: ethanol (fuel), liquid oxygen (oxidizer), and nitrogen gas (pressurant and pneumatics). The main functions of the system include pressurizing the propellants, purging the propellant lines to the engine, venting the tanks, and starting and stopping the flow of propellants to the engine. These functions are controlled by 3-piece PTFE stainless steel ball valves from Assured Automation and are actuated by quarter turn pneumatic actuators, for a total of 8 valves across the two propellant systems. This allows for remote control of these functions and for mechanical spring return to a safe state in case pneumatics or electricity is lost to the system. Stainless steel 316 tubing of 3/8" diameter and 0.035" wall is used for all the lines, except for the pneumatics, which uses 1/4" polyurethane tubing. This was the minimum line size that allowed for the target flow rate without excessive pressure drops and fluid velocities inside the line, and allows for interchangeable fittings and components between the systems. Across the high pressure system, relief valves set to 450 psi opening pressure are used to prevent overpressure. A simplified version of the piping and instrumentation (P&ID) diagram of the system is shown in Figure 7. A complete P&ID can be found in Appendix C.

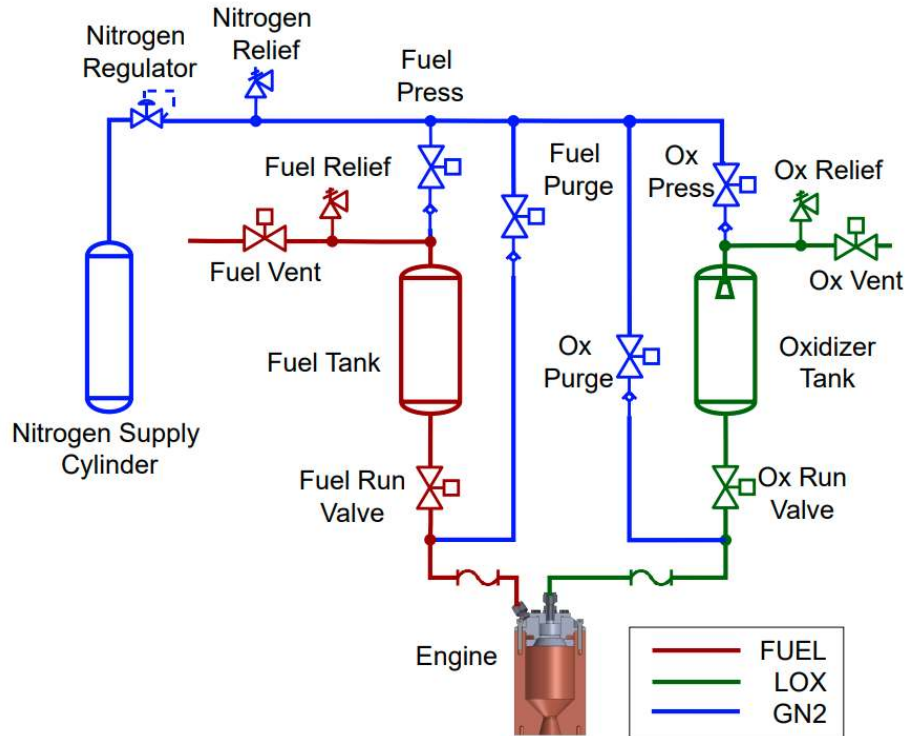


Figure 7: Simplified P&ID diagram

The nitrogen system includes a high pressure circuit (> 100 psi) and a pneumatics circuit (< 100 psi). The nitrogen is supplied from the high pressure regulator on a supply cylinder and enters the high pressure nitrogen circuit, which is used for tank pressurization and purging. This pressure is regulated down to 100 psi for pneumatics. The pneumatics circuit is divided into two separate sections depending on which valves they enable: non-hazardous, and hazardous. Non-hazardous valves are deemed valves that are safe to actuate with operators present but can be depressurized with a rope-actuated valve in an emergency situation. Hazardous valves are valves that may endanger operators near the system if actuated, and are locked out using a manual valve that vents the hazardous pneumatics downstream of it. During a test, operators pressurize the hazardous pneumatics as the last step before retreating from the test area.

The cryogenic and oxidizing nature of liquid oxygen introduces several challenges for the liquid oxygen system, the most significant being boiloff. All of the fluid system components, such as the tank, run valve, and lines need to be chilled down to liquid oxygen's boiling temperature to prevent excessive boiling from causing poor flow conditions during run. This chill in is done using the liquid oxygen itself. To reduce heat transfer from the ambient environment, the tank and long runs of lines are insulated using reflective bubble foil as a material that does not absorb moisture. The seams and the ends of the foil are taped down using aluminum tape to prevent moisture from being trapped under the insulation, which will severely increase boiloff. Another problem is trapping

cryogenic liquid in volumes that have the potential to be completely sealed off, causing the boiling liquid to overpressurize the section. This is solved by adding pressure relief valves on riser tubes that can vent potential dead volumes and drilling relief holes in the balls of the ball valves to ensure liquids always have a path to escape the ball.

9.4 Gimbal

The gimbal acts as a pivot with two degrees of freedom which supports the engine and allows it to be rotated to change the direction of its thrust. It is strong enough to withstand the firing force of the engine while allowing for the thrust vector control actuators to manipulate the engine to a desired angle. During firing the maximum desired angle in any direction is 7 degrees, and the gimbal has hard stops at 10 degrees in case of actuator failure.

A ring style gimbal was selected as it separates the two pivot axes, simplifying manufacturing, and allows for the pass-through of fluid and electrical connections to the engine. An image of the gimbal assembled onto the engine is shown in Figure 9, along with a labelled picture of its components in Figure 8. The gimbal also has two mounted encoders, one for each axis of rotation. By placing the encoders directly onto the axis of rotation instead of measuring the actuator extensions, the angle can be directly measured instead of inferred from the actuator positions. This also eliminates any measurement errors due to manufacturing errors or tolerance stack-up.

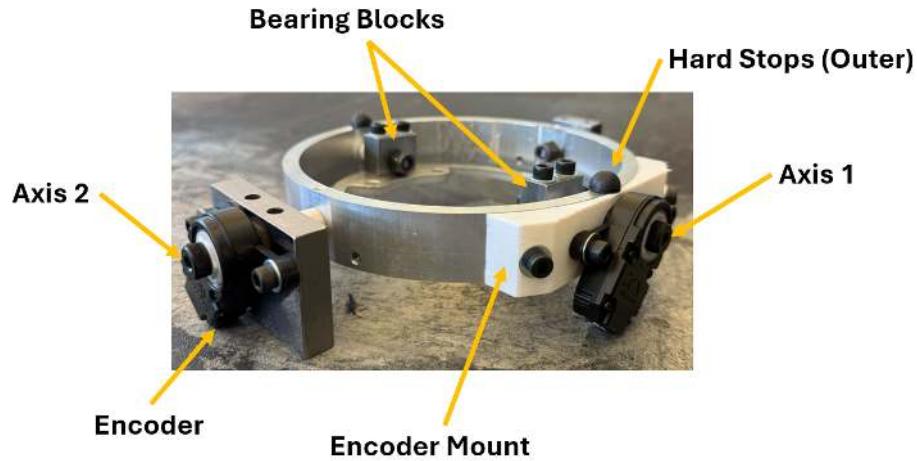


Figure 8: Labeled gimbal.

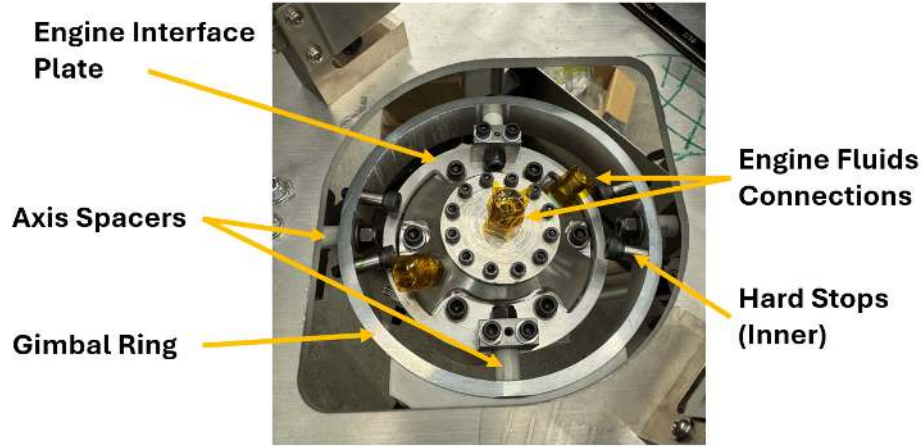


Figure 9: Labeled gimbal with engine mounted.

The two main axes of the gimbal are shoulder bolts sized to withstand the thrust of the engine with a FOS of 2. The shoulder bolts run in bronze oil impregnated bushings (not shown) which allow for smooth operation of the gimbal. The outer axis connects the bearing blocks mounted to the static thrust stand to a gimbal ring. The inner axis is positioned at 90 degrees from the outer axis and connects the gimbal ring to the engine interface plate. Since the axes are shoulder bolts, a 3D printed spacer is placed between the gimbal ring and bearing block to ensure proper spacing and allow for proper tightening which minimizes play in the assembly.

9.5 Thrust Vector Control

The TVC system's main purpose is to move the engine while it is firing in order to change the direction of the produced thrust. On a full flight vehicle, this would provide "steering" which would allow the rocket to change direction and attitude in flight. For the scope of this project, the TVC system consisted of 2 linear actuators that provided motion in 2 axes. The actuators were fixed to a stationary plate on one side to provide a fixed reference and to the bottom of the engine on the other.

The hardware requirements for the thrust vector control actuators (TVCA) are given in Table 2. Other requirements such as accuracy of motion were deemed to be solvable by software and therefore not considered when designing the hardware.

Table 2: TVCA Requirements

Parameter	Value
Angular Speed	0.4 rad/s
Angular Acceleration	2 rad/s ²
Nominal Range of Motion	$\pm 7^\circ$
Maximum Range of Motion	$\pm 10^\circ$

To size the TVCA, a few parameters were determined. These included the extended length, retracted length, range of motion, movement speed and required force. Note that the range of motion refers to the linear range of motion of the actuator between its maximum extension and minimum retraction. This is separate from the angular range of motion of the engine. To get the extended length, retracted length and by extension, the range of motion, a stick model of the system was created in SolidWorks as shown in Figure 10. In this model, key distances such as the distances from the center of rotation to either end of the TVCA mount points were modeled as lines.

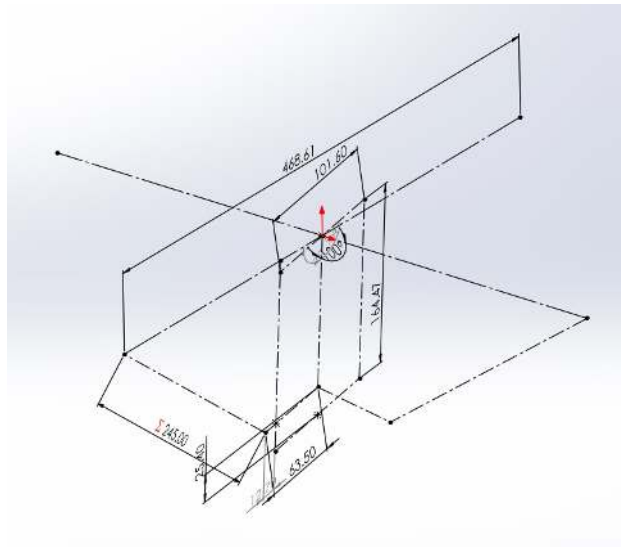


Figure 10: Stick model of the engine, gimbal and TVCA

To determine the extend and retract lengths, additional lines were used to model the length of the TVCA. The engine angle was then varied between $+10^\circ$ to -10° in both axes. At each extremity, the length of the actuators was measured. The maximum and minimum length values were extracted as the minimum extended length and maximum retracted length the TVCA had to be sized for. Combining the extended and retracted length also gave the range of motion. These values are shown in Table 3.

Table 3: TVCA Range of Motion

Parameter	Value
Extended Length	168.84 mm
Retracted Length	197.95 mm
Range of Motion	29.10 mm

To determine the required movement speed, the engine, TVC and gimbal system was simplified into a crank slider mechanism as shown in Figure 24 in Section A.2. The analysis of this system is given in Section A.2 and the result was that the actuators needed a speed of at least $55.96 \frac{\text{mm}}{\text{s}}$.

To determine the force requirement for the TVCA, a free body diagram was drawn of the system as shown in Figure 23 in Section A.1. The equation for calculating the required TVCA force is given by Equation 1 below. Full derivation of the equation is located in Section A.1.

$$F_{\text{TVCA}} = \frac{(I_G + m\bar{r}^2)\alpha + F_e t_1}{\sin \phi(\bar{r} + n) + \cos \phi t_2} \quad (1)$$

The known parameters used in the force calculation are given below in Table 4 with ϕ being a variable based on current engine angle.

Table 4: TVCA Force Inputs

Parameter	Value
I_g	$0.028 \frac{\text{kg}}{\text{m}^2}$
m	8.2 kg
$\bar{r}$	0.0762 m
α	$2 \frac{\text{rad}}{\text{s}^2}$
F_e	500 N
t_1	0.005 m
t_2	0.0508 m
n	0.07874 m

The maximum force exerted by the engine on the TVCA was 21 N or 4.72 lbs. From this, a list of the TVCA parameters can be tabulated in Table 5.

Table 5: TVCA Parameters

Parameter	Value
Extended Length	168.84 mm
Retracted Length	197.95 mm
Range of Motion	29.10 mm
Speed	$55.96 \frac{\text{mm}}{\text{s}}$
Force	21 N

The original intention was to design and build a custom linear actuator to suit the parameters in Table 5. However, off-the-shelf linear actuators are very common, come in a wide variety and would save substantial engineering time. Therefore, it was decided to purchase a linear actuator off the shelf. Nine different linear actuators were selected that met all, if not most of the parameters and were close in others. From those, the selection was narrowed down to the Progressive Automations model PA-01-2-28-N-12VDC [10] and the Pololu Glideforce GF23-120502-1-65 [11]. The final decision was to use the Pololu Glideforce actuator on account of its lower retracted length, higher force and higher speed. The most uncertainty was on the actual force and speed the actuator needed to meet and therefore having a faster, stronger actuator would give more margin for error.

The actuators were mounted to the thrust plate and engine using clevises machined out of aluminum 6061-T6. Since the force on the TVCA was minimal, 6061-T6 was chosen due to its availability and ease of machining. Ball joints were installed on the ends of each actuator to allow them to rotate when the perpendicular actuator was extended or retracted and therefore preventing an over constrained system. The ball joints were mounted to the clevises using clevis pins retained by cotter pins. Washers were used to prevent the ball rod ends from sliding on the clevis pins and thereby introducing error into the control system. The TVCA assembly is shown in Figure 11 and Figure 12. The clevises are attached to the engine and thrust plate using #10-32 bolts with washers and nylon locknuts to prevent disassembly during operation due to vibration. Due to the minimal force acting through the actuators, detailed analysis on the connecting elements was deemed unnecessary due to the low likelihood of failure.



Figure 11: TVCA assembly



Figure 12: Close up of the clevis assembly

The fully integrated TVC system is shown in Figure 13.



Figure 13: TVC system integrated with engine

9.6 Electrical and Software

A printed circuit board (PCB) was built to control the thrust vector control actuators. This board consists of two motor drivers, two battery connectors, a controller area network (CAN) bus transceiver, and a PIC18f26K83 microcontroller. The CAN bus enables two-way communication between the thrust vector control board (TVC-controller) and mission control. It was selected for its robustness and compatibility with the Waterloo Rocketry team's existing infrastructure. The TVC-controller periodically reports the positions of the encoders, the battery voltages, and the current draw of the actuators. The PCB is shown below in Figure 14.



Figure 14: TVCA controller board

The PCB uses a PID loop to control the angle of the engine. The angular position is determined using two encoders mounted to the gimbal. Given these angles, the PID loop determines the correct voltage (modulated using PWM) to supply the actuators to correct any measured error. Operators communicate with the PCB using specialized CAN commands. These commands let the operators change the desired position, initiate a predefined sequence and change various PID parameters.

All of the pneumatic valves in the feed system are controlled by Waterloo Rocketry's Remote Launch Control System (RLCS). The system uses a local area network to send commands from mission control to the valve actuators at the launch pad. This enables tests involving high pressures and flammable fluids to be performed safely at a distance. Originally, RLCS was designed to require every actuation to be specified manually, which does not meet the tight timing requirements of a liquid oxygen engine. As a result, the system was updated such that the operator can run a sequence of predefined actuations. Figure 15 shows the client side of RLCS that houses all the actuator switches and the fire button.



Figure 15: RLCS clientside

In addition to RLCS, this project also uses Waterloo Rocketry' data acquisition system (DAQ). This system contains a National Instrument sensing box, some custom designed PCBs for signal conversion, and a stripped-down Framework laptop. DAQ is used to take measurements from various pressure transducers, load cells, and thermocouples and publish them to the same local area network RLCS uses.

9.7 Ground Support Infrastructure

In addition to the feed system described in Section 9.3, more infrastructure projects were built to support the engine. A test stand was built to provide a structure for the engine and TVC to be secured to. A flame diverter prevents the high-temperature flame from the engine from bouncing back into the engine or damaging the test site. The main structure is welded from steel tubes, with a secondary frame welded from steel L-bars to hold a slab of refractory concrete and 12 firebricks at an angle. The test stand is also an excellent location to mount a sponsorship plate. A steel pan made from bent sheet metal is put below the flame diverter to catch any potential liquid oxygen, stopping it from coming in contact with the asphalt beneath. The entire test stand setup is shown in Figure 16.



Figure 16: Engine Test Stand

Two security cameras and flood lights were installed around the test area for remote monitoring. These are crucial for the operation crews to have a live view of the situation of the engine at a safe distance. The cameras and floodlights, in addition to the RLCS and DAQ systems, are powered by two car batteries instead of generators, minimizing the spark risk near liquid oxygen.

9.8 Safety

The two main categories of risks involved in this project are risks to the public, and risks to the team. Hot fire tests are conducted at the Waterloo Rocketry Team's current test site, which is a small paved area at the end of a publicly-accessible road. Due to the energetic nature of the propellants, a safety radius of about 400 ft around the test stand is enforced during the test. This safety radius is based off the minimum safe distance allowed by the effects of an overpressure wave and flying shrapnel in the case of an explosion where the two propellants mix and combust. The analysis is based on the FAA Calculation of Safety Clear Zones for Experimental Permits under 14 CFR § 437.53(a)[12], which is intended for experimental aerospace testing. It assumes a load of 1.5 kg of ethanol and a full tank of LOX at 12.7 kg, a conservative assumption due to the variability of LOX loading. Figure 17 shows the safety distance at the test site, and calculations of this distance can be found in Appendix B. During a test, this area is closed off from the public with the help of test personnel designated to enforce this perimeter.

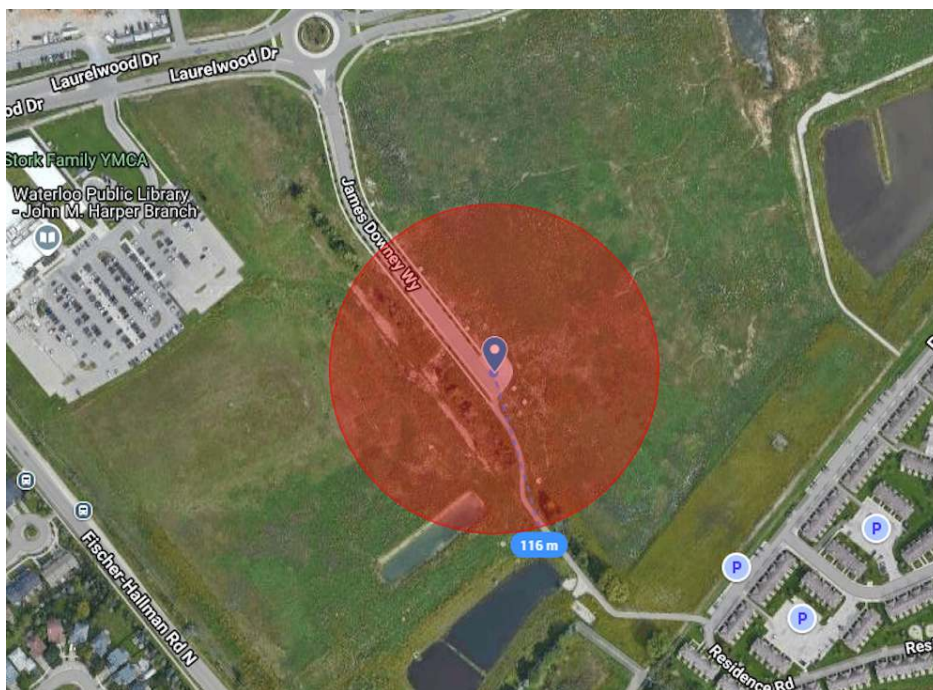


Figure 17: Test site safety radius

The second category of risks are to operators directly involved in the test. These risks include contact with cryogenics, proximity to pressurized gas, and ignition from static discharge due to an oxygen-enriched environment. Firstly, all parts of the fluid system that could potentially come into contact with liquid or gaseous oxygen are precision cleaned using a degreasing solution and thoroughly rinsed and dried to eliminate any grease or dirt that could react with the oxidizer.

Next, manual lock outs and preplanned operations procedures ensure that hazardous sources of pressure can not pressurize the system with operators present. For example, the valves that pres-

surize the propellant tanks with nitrogen have their pneumatics supply vented with operators present (hazardous pneumatics) and so are not able to actuate. These valves are only energized by operators as the last step before they retreat beyond the safety radius for the fully remote portion of the test procedure.

Although the area is well ventilated since testing is performed outdoors, the area surrounding the test stand could become enriched with oxygen due to venting of gaseous or liquid oxygen when operators are present. To minimize the risk of an ignition, all operators are required to wear 100% cotton clothing to minimize static electricity buildup. Furthermore, the person loading the fuel into the system is different from the person loading the liquid oxygen to prevent exposure to the fuel from becoming a possible source of ignition.

Lastly, operators working with liquid oxygen are required to wear face shields and cryogenic aprons and gloves to protect them from any possible splashes of cryogenic liquids or the cold temperatures of the hardware.

10 Results

Many of the systems designed in this project have been successfully validated, although some future work remains. The feed system, TVC system, gimbal, ground support infrastructure and electronics/software have met all of their requirements successfully. While the engine has not yet been static fired successfully, its design has also been validated at the expected pressures, temperatures and flow rates in cold-flow testing.

The first milestone for validating the engine and feedsystem was to ensure the system could function at cryogenic temperatures while maintaining the expected pressures and flow rates. This was achieved through performing “cold flow” tests where ethanol and inert liquid nitrogen were flowed through the feed system and engine. Here liquid nitrogen is used instead of liquid oxygen as it is significantly safer to work with while maintaining similar fluid dynamics and temperature. The first cold flow performed included both the fuel and oxidizer systems, and was able to successfully validate the expected flow rates on the fuel side and that the oxidizer side functioned well at cryogenic temperatures. The fuel flow rate was measured at 145 g/s, extremely close to the predicted value of 154 g/s at 100 psi. Unfortunately there were issues measuring the mass of the oxidizer tank as the mount shifted due to the cryogenic temperatures, and so the oxidizer flow rate could not be determined. The fuel data from this test is shown in Figure 18.

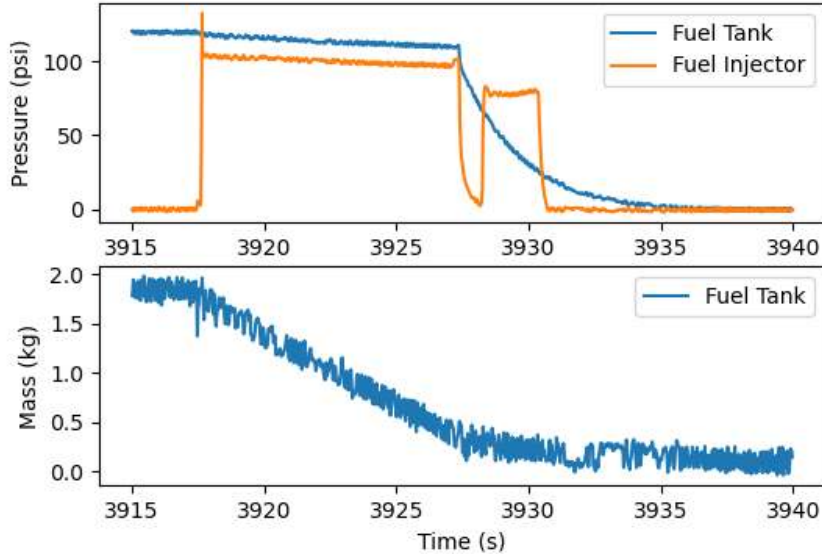


Figure 18: Cold flow 1 fuel data

A second cold flow of just the oxidizer side was performed after modifying the oxidizer tank mass measurement system to decouple it from any thermal effects. This led to obtaining good data, however the measured flow rate of 63 g/s was well below the expected value of 100 g/s at 80 psi. This was attributed to incomplete cooling of the injector elements before performing the cold flow, leading to the oxidizer boiling as it came in contact with the warm injector and thus reducing its flow rate. Based on the two cold flows the feed system was considered fully validated as it was able to provide the required flow rates to the engine. Additionally, as the geometry of the fuel and oxidizer orifices in the injector are identical, it was decided that the orifice parameters as measured by the fuel flow could be applied to the oxidizer side, thus avoiding the need for further cold flows. The oxidizer data from this test is shown in Figure 19.

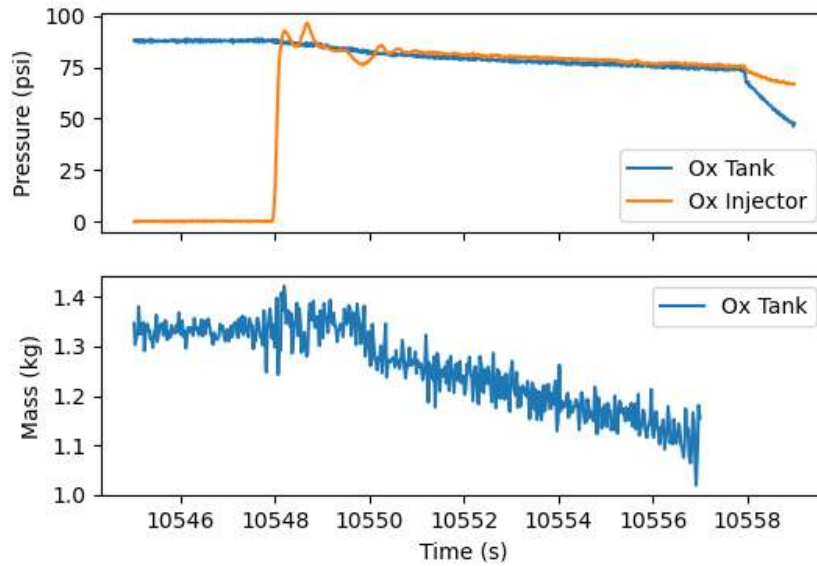


Figure 19: Cold flow 2 oxidizer data

The electronics, software, gimbal and TVC system were first validated through a “wiggle test”. Here a 3D-printed mockup of the engine was mounted to the gimbal and the TVC actuators were attached, as shown in Figure 20. The position of the engine was then able to be successfully controlled, validating the performance of the control electronics and software while exercising the gimbal and TVC system.



Figure 20: TVC wiggle test assembly

The last tests performed so far were two integrated static fire attempts. The goal of a static fire is to fire the engine for a full-duration 10 second burn while it is bolted to the thrust stand in order to measure the force produced, and additionally the demonstrate thrust vector control during the burn. At the Waterloo Rocketry team's test site, the system was fully assembled and a firing was attempted. During the first attempt, an operational issue led to the oxidizer run valve being left open while the oxidizer tank was pressurizing. This caused all of the liquid oxygen to flow out of the engine prematurely, thus aborting the attempt. Despite the abort, data was able to be collected about the oxidizer side flow rate, and it was verified that when properly chilled the oxidizer flow rate matched the predicted value.

A second static fire was then attempted later the same day, resolving the operational issue and leading to an attempt to ignite the engine. Unfortunately the engine did not ignite, likely due to a combination of factors such as tank pressures being too low and the igniter not being sufficiently secured. The igniter was ejected from the engine when the oxidizer and fuel run valves were opened, as can be seen in Figure 21.

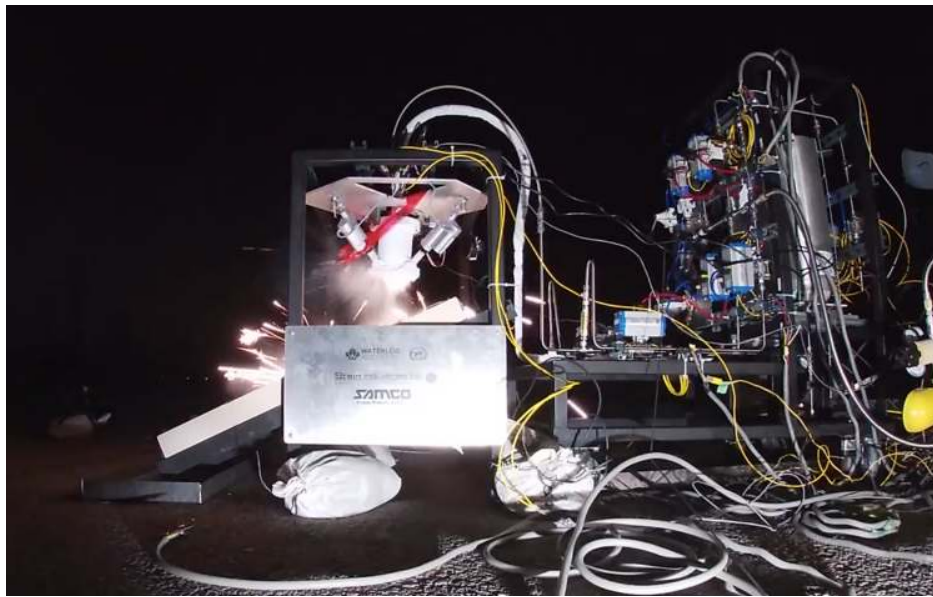


Figure 21: Igniter being ejected during the second static fire attempt

The TVC system and gimbal were further validated during this test, as a sequence of moves were performed while the propellants were flowing out of the engine. Figure 22 shows the data gathered from the encoders mounted to the gimbal. This directly shows the performance of the TVCA since gimbal motion is controlled by the TVCA.

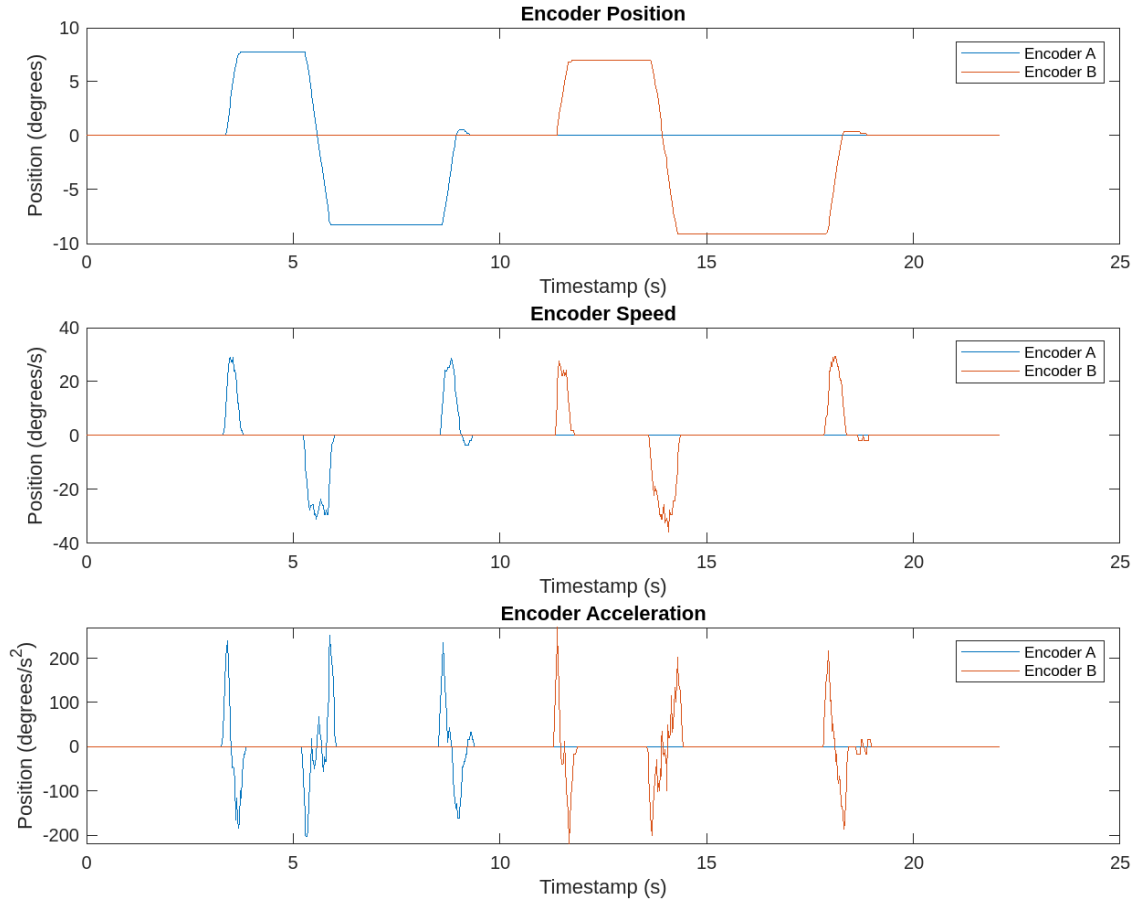


Figure 22: Position, Speed and Acceleration of the TVCA

The key performance parameters are given in Table 6.

Table 6: TVCA Performance Parameters

Parameter	Value
Maximum Rotation Angle of Gimbal	7.73°
Minimum Rotation Angle of Gimbal	-8.79°
Maximum Angular Velocity	29.24 °/s
Minimum Angular Velocity	-35.89 °/s
Maximum Angular Acceleration	269.53 °/s ²
Minimum Angular Acceleration	-220.50 °/s ²

11 Discussion

While an integrated static fire test has not yet been achieved, all of the systems designed as part of this project have successfully completed various tests of their design and requirements. These designs, along with the associated hardware, are all available to members of the Waterloo Rocketry

design team in addition to being open source where applicable. Discussions with the team relating to the future of the designs proven in this project are ongoing, and it is likely that they will continue to be built upon as desired.

One of the key learnings from this project has been how to successfully design systems to work with cryogenic liquid oxygen. The feed system functioned exactly as designed and was able to fill and supply liquid oxygen to the engine at the desired parameters. The engine was able to contain flowing cryogenics and the measured flow rates aligned with expected values from the design. Additionally, issues that likely led to the failed ignition have been identified and another static fire attempt is planned in the coming week, aiming to fully demonstrate the successful engine design.

The thrust vector control system has met all of its performance parameters when comparing Table 2 to Table 6. The range of motion is greater than 7° but less than 10° in both directions. The speed of the actuators is at least $29.24^\circ/\text{s}$ which is around 0.51rad/s and greater than the requirement of 0.4rad/s . The acceleration of the actuators is at least $220.50^\circ/\text{s}^2$ which corresponds to 3.85rad/s^2 and is greater than the required 2rad/s^2 . Based on these results, the thrust vector control system is sufficient to control a small scale hopper vehicle during flight.

12 Conclusion

Although a successful integrated static fire test was not achieved, this project has significantly advanced Waterloo Rocketry in its understanding of cryogenic propellants and thrust vector control. A total of four tests were successfully completed using liquid nitrogen and liquid oxygen. With extensive research and the expertise of an industry professional, all of these tests were completed without compromising the safety of personnel or equipment. Excellent data was collected from these tests which validated the design of the engine and the thrust vector control system independently. Several recommendations for future liquid oxygen and ethanol powered rocket engines resulted from the operational challenges and ignition failure experienced in the integrated static fire attempts. The hardware used in this project will be valuable to Waterloo Rocketry for further development of thrust vector control and cryogen propelled rockets. Future work could fully validate this ground system and proceed to optimize and flight-weight all systems for an integrated hopper vehicle.

13 Recommendations

The following are recommendations to the Waterloo Rocketry student design team based on the lessons learned during this project.

On the use of thrust vector control, it has been proven that thrust vector control is possible. Further development and ground testing is encouraged, however the use of thrust vector control is not currently permitted on a flight vehicle by the Launch Canada competition [13]. Therefore, it

is recommended that if the TVC system development continues, parallel efforts should be started to obtain regulatory approval for flight tests of such a vehicle.

On the use of cryogenic propellants such as liquid oxygen, further study is required on whether the added resources required to support cryogenic propellant use is feasible for student rocketry. It is highly recommended that all future work on cryogenics follow the safety processes outlined in Section 9.8 and further safety development outlined in Section 14.

Developing compact, mobile test infrastructure is highly recommended due to its ease of use and transport. This may not be feasible for engines above a certain size. In absence of this, it is recommended that infrastructure that only requires assembly once be prioritized.

For most purposes, heatsink engines are not recommended due to their extreme weight and potential costs. The copper engine used during the project cost over \$300 and only allowed for a 10 second burn at an O/F ratio specifically tailored to reduce temperature.

14 Future Work

The scope of this project was to build a proof-of-concept ground test engine to lay the groundwork for the use of cryogenic propellants and thrust vector control. The following are ideas for future work beyond the scope of this project.

The main issue with the engine is that it is made out of a large and extremely heavy round of copper. This was because the engine was designed to absorb the heat of combustion and the O/F ratio was deliberately made less efficient to reduce the heat generation. One of the most apparent challenges that warrants future work is how to decrease the weight of the engine while still allowing for increased burn time. This could include regenerative cooling, graphite inserts or the use of ablative materials which warrant less material than a heatsink engine. Increasing burn duration is crucial to developing a full flight hopper vehicle. The engine was designed for a burn time of 10 seconds to give adequate time to test the TVC system. For a flight hopper vehicle, this burn time would likely need to be extended by at least 2-3 times. Work could also be done on improving performance, however this is secondary to burn time as the purpose of a hopper is to hop/hover and not fly high. The injectors would need to be examined for potential optimizations with the use of liquid oxygen or other cryogenic propellants. A long term project couple be to integrate a pump system into the engine which would massively increase propellant flow rates and pressures.

Supporting the engine is the feed system which itself would need to be improved along with the engine. As larger flow rates are required, the feed system would need to accommodate higher tank pressures and higher flow rates. This would include increasing the size of plumbing and actuation power of valves. To gather more data on flow rate and mass, especially on the cryogenic side, the feed system would need better load cells or other accurate form of fill sensing. This could include the implementation of a venturi flow meter.

The thrust vector control and gimbal system can be further made flight-like through light-weighting and shrinking. Currently, the footprint of the TVC and gimbal system is larger than the footprint of the engine. To make the system flight-like, work should be put into reducing the size of these systems such that they are smaller than the footprint of the engine. This is especially true of the actuators which are overbuilt for their purpose and are therefore heavier and larger than they need to be. Work should also be placed on reducing the weight of components.

Future work on ground support infrastructure can be split into test infrastructure for supporting testing and flight infrastructure for supporting a hop. Further characterization of the system is needed which would involve more accurate load cells that can measure forces and moments in more than 1 axis. Increasing the size and thrust of the engine will require a larger, more robust test stand and new methods to keep the system on the ground during testing. A flight vehicle would require a new test site capable of supporting a hop. This includes concrete pads at the takeoff and landing zone, a large area clear of obstructions, adequate blast shielding and regulatory approval.

On the software and electrical side, the control algorithms would need to be improved to support a full hopper vehicle. This includes testing the balancing of a rocket-like mass like an inverted pendulum to see if the TVC system is capable of the true speeds and accelerations required for flight. With a more powerful engine and better actuators, the electrical system that powers the TVCA would need to be capable of providing the increased power required. The power source would also need to be small and light enough to fly on a hopper vehicle.

There are a few safety measures that can also be improved. There is currently no good storage solution for cryogenics and specifically liquid oxygen. If cryogenics are to be pursued in a serious capacity in the future, a dedicated storage facility should be set up and must include the following at minimum:

- A fenced or closed off area such that only trained and approved personnel may enter
- Adequate ventilation to prevent gas buildup and a way to rapidly cycle air to vent gas buildup
- Gas monitoring to alert personnel of any elevated levels of potentially reactive, flammable or hazardous gas or to the lack of adequate breathing air
- The ability to enter and exit the storage area without the risk of creating sparks
- Proper signage indicating the type of gas stored and risks, in accordance with University policies

It must be stressed that any future work be done with an emphasis on safety. This means working with the proper authorities and experts, obtaining proper regulatory approval for tests, following strict test plans, considering all risks and training personnel.

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A TVCA Calculations

A.1 Force

To start off, a free-body diagram of the system is shown in Figure 23.

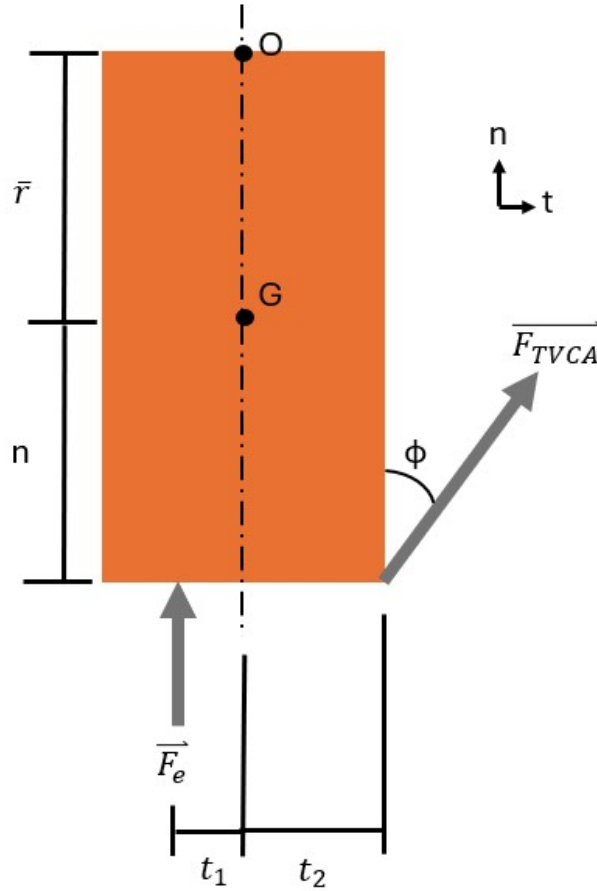


Figure 23: Free body diagram of the engine

- O is the pivot point (center of rotation) of the engine
- G is the center of mass of the engine
- $\bar{r}$ is the vertical distance between the pivot point (center of rotation) and the center of mass of the engine
- n is the vertical distance between the center of mass of the engine and the TVCA mount point
- t_1 is the horizontal distance between the center line of the engine and the thrust force of the engine

- t_2 is the horizontal distance between the center of mass of the engine and the TVCA mount point
- $\vec{F}_e$ is the force of thrust from the engine
- $\vec{F}_{TVCA}$ is the force from the TVCA on the engine
- The normal and tangential coordinate directions are in reference to the bottom plane of the engine

From Newton's Second Law, the net moment is equal to angular acceleration multiplied by the moment of inertia about the center of mass (point G).

$$\Sigma M_G = I_G \alpha$$

To get the rotation about the engine pivot point (point O), use the Parallel Axis Theorem.

$$\Sigma M_o = (I_G + m\bar{r}^2)\alpha$$

The external moments acting on the system are the force of the engine at a distance of t_1 and the TVCA force. The TVCA force can be split up into normal and tangential components. The force component in the normal direction is $\vec{F}_{TVCA} \cos \phi$ acting at a distance of t_2 . The force component in the tangential direction is $\vec{F}_{TVCA} \sin \phi$ acting at a distance of $\bar{r} + n$. Substituting those into the equation and considering a counterclockwise moment as positive gives

$$\vec{F}_{TVCA} \cos \phi(t_2) + \vec{F}_{TVCA} \sin \phi(\bar{r} + n) - F_e t_1 = (I_G + m\bar{r}^2)\alpha$$

Finally, rearranging and isolating for the TVCA force gives

$$\begin{aligned} \vec{F}_{TVCA} \cos \phi(t_2) + \vec{F}_{TVCA} \sin \phi(\bar{r} + n) &= (I_G + m\bar{r}^2)\alpha + F_e t_1 \\ \vec{F}_{TVCA} &= \frac{(I_G + m\bar{r}^2)\alpha + F_e t_1}{\sin \phi(\bar{r} + n) + \cos \phi(t_2)} \end{aligned}$$

A.2 Speed

To calculate the actuation speed of the TVCA, the system was approximated as a crank-slider linkage mechanism as shown in Figure 24.

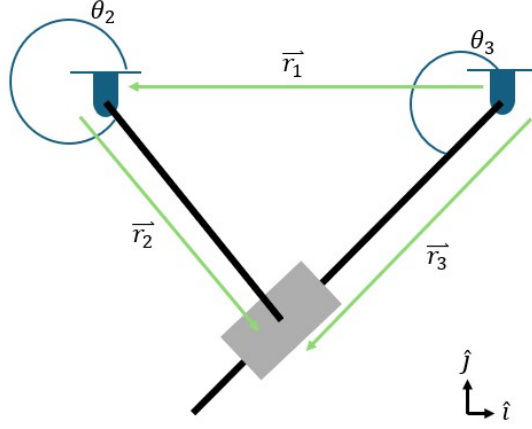


Figure 24: TVC linkage model

- $\vec{r}_1$ represents the distance from the outer gimbal mounting position to the center of the gimbal (center of rotation of the engine)
- $\vec{r}_2$ represents the distance from the center of rotation of the engine to the TVCA mounting point
- $\vec{r}_3$ represents the distance between the TVCA mounting point on the engine and its mounting point on the thrust plate
- $\vec{\theta}_2$ represents the angle from the center of rotation to the TVCA mount point added to the current rotation angle of the engine
- $\vec{\theta}_3$ represents the angle of the TVCA relative to the thrust plate
- The slider represents the ability of the TVCA to retract and extend along $\vec{r}_3$
- The directions of the r vectors are arbitrary

The actuation speed of the TVCA is represented by the velocity of the slider or in other words, $\dot{r}_3$. To find $\dot{r}_3$, first the relationship between the r vectors can be written as

$$\begin{aligned}\vec{r}_1 + \vec{r}_2 - \vec{r}_3 &= 0 \\ r_1 \hat{i} + r_2(\cos \theta_2 \hat{i} + \sin \theta_2 \hat{j}) &= r_3(\cos \theta_3 \hat{i} + \sin \theta_3 \hat{j})\end{aligned}$$

Splitting the equations up into $\hat{i}$ and $\hat{j}$ components yields

$$\hat{i} : r_1 + r_2 \cos \theta_2 = r_3 \cos \theta_3 \quad (2)$$

$$\hat{j} : r_2 \sin \theta_2 = r_3 \sin \theta_3 \quad (3)$$

Taking the derivative of the Equation 2 and Equation 3 with respect to time gives.

$$\dot{r}_1 + \dot{r}_2 \cos \theta_2 - r_2 \dot{\theta}_2 \sin \theta_2 = \dot{r}_3 \cos \theta_3 - r_3 \dot{\theta}_3 \sin \theta_3 \quad (4)$$

$$\dot{r}_2 \sin \theta_2 + r_2 \dot{\theta}_2 \cos \theta_2 = \dot{r}_3 \sin \theta_3 + r_3 \dot{\theta}_3 \cos \theta_3 \quad (5)$$

r_1 and r_2 are fixed lengths so they both equal 0. Equation 4 and Equation 5 become

$$-r_2 \dot{\theta}_2 \sin \theta_2 = \dot{r}_3 \cos \theta_3 - r_3 \dot{\theta}_3 \sin \theta_3 \quad (6)$$

$$r_2 \dot{\theta}_2 \cos \theta_2 = \dot{r}_3 \sin \theta_3 + r_3 \dot{\theta}_3 \cos \theta_3 \quad (7)$$

By multiplying Equation 6 by $\cos \theta_3$ and Equation 7 by $\sin \theta_3$ and then adding the 2 equations together, an expression for $\dot{r}_3$ can be obtained.

$$\begin{aligned} -r_2 \dot{\theta}_2 \sin \theta_2 \cos \theta_3 + r_2 \dot{\theta}_2 \cos \theta_2 \sin \theta_3 &= \dot{r}_3 \cos \theta_3^2 - r_3 \dot{\theta}_3 \sin \theta_3 \cos \theta_3 + \dot{r}_3 \sin \theta_3^2 + r_3 \dot{\theta}_3 \cos \theta_3 \sin \theta_3 \\ r_2 \dot{\theta}_2 (\cos \theta_2 \sin \theta_3 - \sin \theta_2 \cos \theta_3) &= \dot{r}_3 (\cos \theta_3^2 + \sin \theta_3^2) \\ r_2 \dot{\theta}_2 \sin(\theta_3 - \theta_2) &= \dot{r}_3 \end{aligned}$$

The only unknown in the equation is θ_3 which can be found by dividing Equation 3 by Equation 2.

$$\begin{aligned} \frac{r_2 \sin \theta_2}{r_1 + r_2 \cos \theta_2} &= \frac{r_3 \sin \theta_3}{r_3 \cos \theta_3} \\ \frac{r_2 \sin \theta_2}{r_1 + r_2 \cos \theta_2} &= \tan(\theta_3) \\ \tan^{-1} \frac{r_2 \sin \theta_2}{r_1 + r_2 \cos \theta_2} &= \theta_3 \end{aligned}$$

B Safety Radius Calculations

Table 7: Safety Radius Calculations

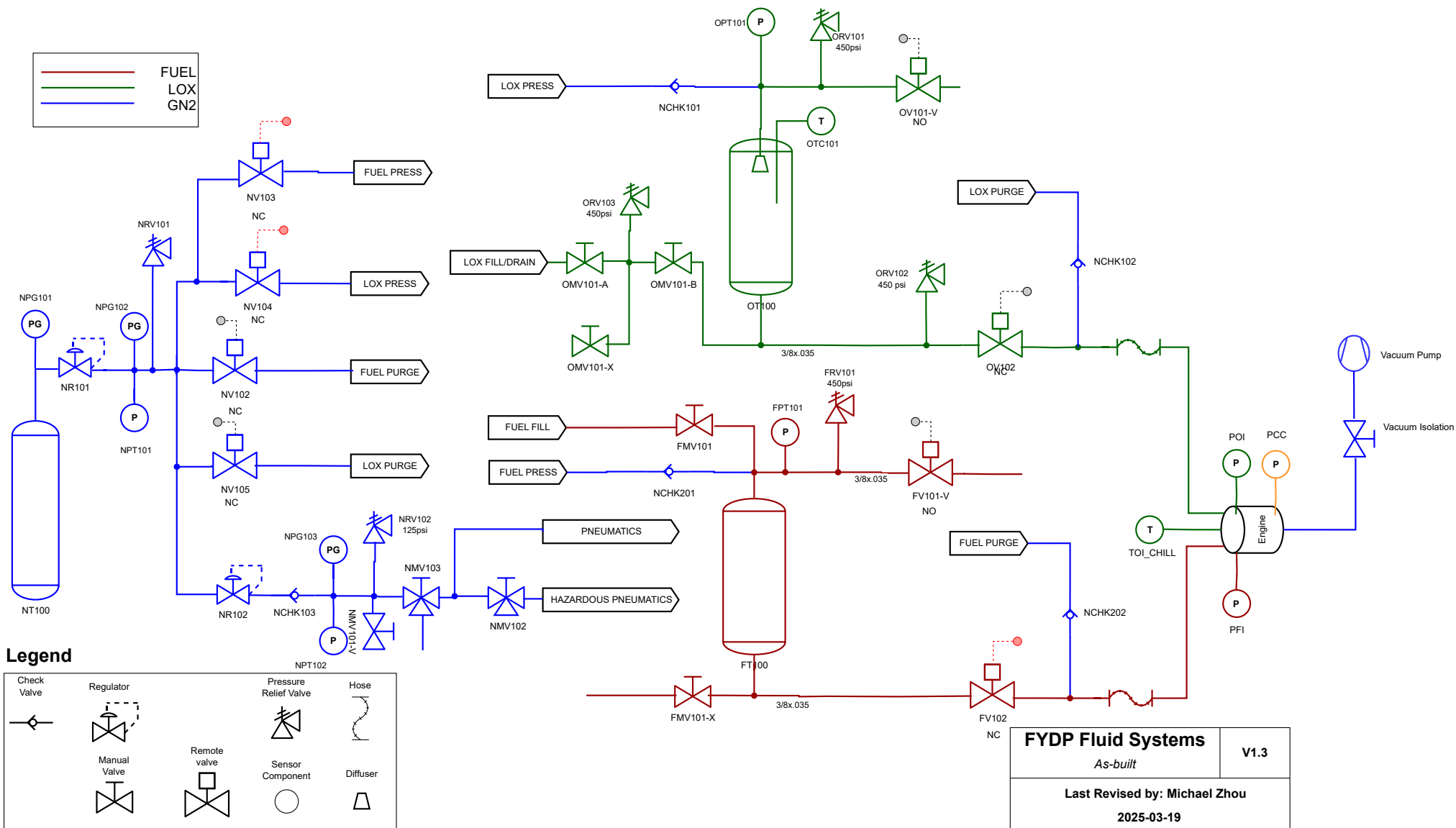
Parameter	Value	Units
LOX mass	12.7	kg
Fuel mass	1.5	kg
Total propellant mass	14.2	kg
Net Explosive Weight factor ¹	0.1	
Net explosive weight	1.42	kg
	3.12	lbm
Overpressure distance ²	65.8	ft
Hazardous fragment distance ³	381.5	ft

¹Appendix E to Part 420, Table E-1, Substitute Alcohol for RP-1

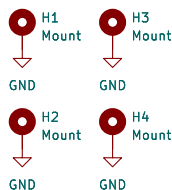
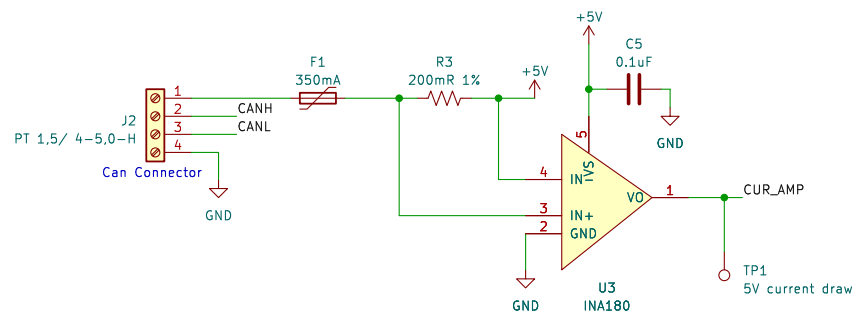
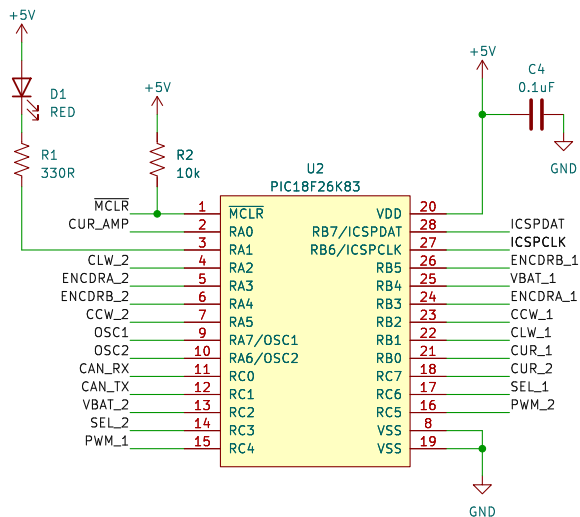
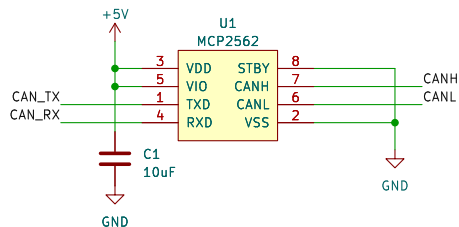
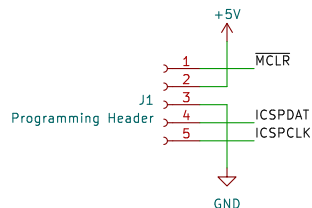
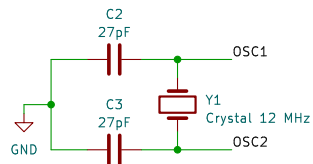
²K factor for Hazard Division 1.1 per propellant combination

³Appendix E to Part 420, 0.5 lbs <NEW<100 lbs

C P&ID Diagram of the Feed System



D TVCA Controller Board Schematics



Waterloo Rocketry

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File: tvca_controller.kicad_sch

Title: T.V.C.A. Controller

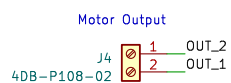
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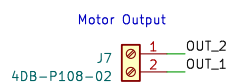
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